

Fault Tolerant Mechanisms

Hardware

TMR

Duplex

Standby
Sparing

Lockstep

Software

N-Version
Programming

Recovery
Blocks

Self-
Checking

Design
Diversity

Data

Hamming

BCH

Reed-
Solomon

LDPC

Time

Repeated
Execution

Roll-Back

Shadow
Latch

Delayed
Clock

Hybrid

Multi-Layer

Hamming +
TMR

DPR + TMR

Lockstep +
TMR